

NOVO SELO, Bulgaria

WMO: 155010

Lat: 44.160N

Lon: 22.785E

Elev: 38

StdP: 100.87

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-9.0	-13.9	1.1	-11.5	-10.5	1.5	-8.4	14.5	5.4	12.4	3.9	1.3	50	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	33.8	23.5	32.3	23.2	30.8	22.6	24.7	31.8	23.9	31.1	23.1	29.8	2.5	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.1	29.6	21.6	16.3	28.7	20.7	15.5	27.4	75.1	31.9	71.9	31.3	68.5	29.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.3	10.3	9.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-15.0	36.0	4.6	2.3	-18.3	37.6	-21.0	39.0	-23.6	40.2	-27.0	41.9		
			-15.1	26.5	4.8	1.2	-18.5	27.4	-21.3	28.1	-24.0	28.8	-27.5	29.7		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	0.2	1.6	7.0	12.9	18.1	21.8	24.3	23.7	18.9	12.3	6.4	1.4	
	DBStd	9.40	5.06	5.35	4.56	2.98	3.23	3.18	2.60	2.56	3.24	3.93	3.87	4.00	
	HDD10.0	1072	305	236	112	8	0	0	0	0	0	22	120	268	
	HDD18.3	2705	562	468	351	163	45	7	1	2	32	191	358	525	
	CDD10.0	1967	1	1	19	97	252	356	443	424	268	93	12	1	
	CDD18.3	559	0	0	0	2	39	113	186	168	49	3	0	0	
	CDH23.3	5284	0	0	1	19	312	1071	1897	1562	396	26	0	0	
	CDH26.7	2021	0	0	0	2	54	388	833	637	106	2	0	0	
Wind	WSAvg	2.5	2.3	2.6	3.2	2.5	2.7	2.6	2.5	2.5	2.2	2.2	2.3		
Precipitation	PrecAvg	556	37	40	34	47	55	54	50	44	42	45	37	62	
	PrecMax	794	64	99	72	77	119	166	120	156	105	155	97	140	
	PrecMin	312	6	1	4	0	25	13	8	0	3	3	0	20	
	PrecStd	125	19	29	22	21	25	37	32	41	33	40	27	37	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	15.8	21.8	26.0	30.1	34.0	37.6	35.5	31.7	26.5	18.7	14.2	
		MCWB	9.1	10.3	12.5	17.4	20.7	24.0	23.8	23.6	21.1	18.7	13.7	10.1	
	2%	DB	10.7	13.1	18.7	22.6	28.2	32.2	34.6	33.4	29.5	23.0	15.7	9.8	
		MCWB	7.4	9.2	12.1	15.0	20.4	23.6	23.6	23.6	20.7	17.4	12.4	7.6	
	5%	DB	8.3	10.7	16.1	20.6	26.7	30.6	32.6	32.0	27.6	20.5	13.7	7.8	
		MCWB	5.9	7.7	10.8	14.2	19.6	22.7	23.3	22.9	19.8	16.2	11.4	6.1	
	10%	DB	6.2	8.3	13.9	18.9	25.1	28.9	30.9	30.3	25.6	18.4	12.0	6.3	
		MCWB	4.6	6.2	9.6	13.5	18.9	21.8	22.9	22.2	18.8	15.0	10.4	5.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	11.0	13.7	18.1	22.2	25.3	25.5	25.7	22.3	19.8	14.6	10.3
			MCDB	13.5	14.6	19.9	24.8	27.7	32.1	33.1	32.3	28.9	24.6	17.7	13.5
2%		WB	7.5	9.3	12.3	15.8	21.0	24.2	24.7	24.6	21.2	18.1	13.1	8.2	
		MCDB	10.4	12.7	17.9	20.9	27.1	30.8	32.1	31.5	28.4	21.8	14.9	9.6	
5%		WB	6.1	7.8	11.1	14.8	20.1	23.1	24.0	23.6	20.2	16.6	11.8	6.5	
		MCDB	8.0	10.3	15.7	19.5	26.0	29.6	31.6	30.6	26.7	20.2	13.3	7.5	
10%		WB	4.9	6.3	9.8	13.8	19.2	22.2	23.2	22.7	19.3	15.1	10.5	5.3	
		MCDB	6.1	8.2	13.4	18.3	24.5	28.4	30.0	29.3	24.7	18.2	11.9	6.3	
Mean Daily Temperature Range	5% DB	MDBR	5.3	6.1	8.4	9.5	10.3	11.1	11.8	11.5	10.5	8.2	6.0	4.7	
		MCDBR	8.8	10.3	12.5	12.9	13.1	13.8	14.6	14.1	13.9	11.7	8.7	7.1	
		MCWBR	5.9	6.9	7.3	7.1	6.7	6.5	6.8	6.7	7.2	7.1	5.7	5.2	
	5% WB	MCDBR	8.0	10.1	11.8	11.4	12.3	13.0	13.7	13.0	12.8	10.7	7.6	5.9	
		MCWBR	5.6	6.9	7.3	6.9	6.8	6.4	6.7	6.8	7.0	6.9	5.2	4.7	
Clear-Sky Solar Irradiance	taub	0.346	0.374	0.412	0.472	0.453	0.446	0.459	0.476	0.445	0.404	0.373	0.346		
	taud	2.283	2.199	2.173	2.026	2.148	2.207	2.177	2.123	2.173	2.281	2.312	2.311		
	Ebn at Noon	752	795	815	792	823	829	813	781	771	754	714	712		
	Edh at Noon	86	111	130	163	148	141	143	146	129	102	84	77		
All-Sky Solar Radiation	RadAvg	1.39	2.20	3.39	4.58	5.59	6.30	6.33	5.63	4.01	2.61	1.50	1.18		
	RadStd	0.20	0.32	0.44	0.52	0.48	0.53	0.42	0.50	0.35	0.37	0.21	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	+0.60	N/A	N/A	N/A	+0.38	N/A	N/A	-153	+101	N/A

Nomenclature: See separate page